

Historical partial packet: unital subspaces of $C(K)$

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Historical Status

This note records the first partial result obtained for the tensor-factor descent question following Proposition 2.8 of Aron–Choi–Kim–Lee–Martin, arXiv:1305.6420. It is superseded by the parent packet, which proves the qualitative result for every nonzero tensor factor E .

Theorem

Theorem 1. *Let K be a compact Hausdorff space and let E be a closed linear subspace of $C(K)$ containing 1_K . Let X and Y be Banach spaces. If $(X, E \widehat{\otimes}_\varepsilon Y)$ has the BPBp with function η , then (X, Y) has the BPBp with the same function η .*

Proof

By injectivity of the injective tensor norm, the inclusion $E \hookrightarrow C(K)$ induces an isometric embedding

$$E \widehat{\otimes}_\varepsilon Y \hookrightarrow C(K) \widehat{\otimes}_\varepsilon Y \subset C(K, Y).$$

Fix $\varepsilon > 0$. Let $T \in S_{\mathcal{L}(X, Y)}$ and $x_0 \in S_X$ satisfy $\|Tx_0\| > 1 - \eta(\varepsilon)$. Define

$$J : Y \rightarrow E \widehat{\otimes}_\varepsilon Y, \quad J(y) = 1_K \otimes y,$$

and set $\tilde{T} = J \circ T$. Since J is an isometry, $\tilde{T}$ is norm one and $\|\tilde{T}x_0\| = \|Tx_0\| > 1 - \eta(\varepsilon)$.

Apply the BPBp for $(X, E \widehat{\otimes}_\varepsilon Y)$. There exist $\tilde{S} \in \mathcal{L}(X, E \widehat{\otimes}_\varepsilon Y)$ and $x_1 \in S_X$ such that

$$\|\tilde{S}\| = \|\tilde{S}x_1\| = 1, \quad \|\tilde{S} - \tilde{T}\| < \varepsilon, \quad \|x_1 - x_0\| < \varepsilon.$$

Viewed in $C(K, Y)$, the function $\tilde{S}x_1$ attains its norm at some $t_1 \in K$. Let Q_{t_1} be evaluation at t_1 , and set

$$S = Q_{t_1} \circ \tilde{S}.$$

Then $\|S\| \leq 1$ and

$$\|Sx_1\| = \|(\tilde{S}x_1)(t_1)\| = 1,$$

so S attains its norm at x_1 . Finally, $Q_{t_1}J = \text{Id}_Y$, hence

$$\|S - T\| \leq \|\tilde{S} - \tilde{T}\| < \varepsilon.$$

Thus (X, Y) has the BPBp with the same modulus.

Supersession Note

The parent full packet replaces the concrete unital $C(K)$ -subspace hypothesis by no hypothesis beyond $E \neq 0$, at the cost of replacing $\eta(\varepsilon)$ by $\eta(\varepsilon/2)$. The theorem above remains as a same-modulus corollary.